Yogendra Sangoju

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Summary

- Quantitative analyst and ML engineer with 2+ years of experience building **financial models, data pipelines, and analytics tools** across equity research, M&A, and capital markets. Strong foundation in **financial engineering** — spanning Monte Carlo simulation, time-series forecasting, scenario analysis, and hedging strategy optimization — paired with production-grade Python and SQL. Proven ability to turn large, complex datasets into decision-ready insights for research and trading stakeholders. Targeting **quantitative finance and capital markets** roles where rigorous modeling meets deal execution.

Experience

- **Quantitative Research Analyst** Jan 2024 – Dec 2024
  Wells Fargo | New York, NY
  - Built **quantitative models** in Python to analyze pricing dynamics across **equity and fixed income markets** — covering historical trend decomposition, factor attribution, and sensitivity analysis under varying rate environments.
  - Conducted **transaction-level data analysis** on large market activity datasets, surfacing patterns that directly informed research recommendations distributed to senior stakeholders.
  - Engineered end-to-end **data ingestion and transformation pipelines** (Python, SQL) to process structured and alternative datasets at scale, reducing manual preparation time by ∼40%.
  - Applied **scenario analysis and Monte Carlo simulation** to stress-test model assumptions and quantify tail risk across different market conditions.
  - Designed **interactive dashboards** (Power BI, Streamlit) used daily by research and trading desks to track portfolio metrics, flag anomalies, and support client-facing reporting.
  - Improved **data quality standards and reporting workflows** in collaboration with cross-functional teams, cutting turnaround time by 2 days per reporting cycle.
- **Machine Learning Engineer** Mar 2023 – Dec 2023
  Optum (UnitedHealth Group) | Hyderabad, India
  - Developed **predictive models** (XGBoost, Random Forest, logistic regression) on large-scale structured datasets, achieving a 22% improvement in out-of-sample accuracy through rigorous feature engineering and hyperparameter tuning.
  - Designed and maintained **high-throughput data pipelines** using Python (Pandas, NumPy) and SQL, handling millions of records daily with automated validation and error logging.
  - Deployed **ML inference services via FastAPI, Docker, and AWS**, enabling real-time prediction APIs consumed by downstream analytics teams.
  - Implemented **LLM-based retrieval pipelines** using FAISS vector search, cutting analyst query time on internal knowledge bases by ∼60%.

Projects

- **M&A Deal Scoring & Transaction Analytics Engine**
  - Built a **quantitative deal-scoring framework** to analyze 10,000+ M&A transactions — computing EV/EBITDA multiples, comparable company valuations, and deal premium distributions across industries.
  - Trained **XGBoost and logistic regression classifiers** to predict deal closure probability (AUC 0.84), with SHAP-based feature attribution identifying key drivers: leverage ratio, sector, and acquirer size.

- Engineered data normalization and enrichment pipelines integrating financial datasets; built interactive dashboards enabling filtering by sector, deal size, and geography.

- **Equity Issuance Pricing & Hedging Strategy Optimizer**
  - Developed **financial engineering models for equity issuance pricing** — incorporating comparable company analysis, book-building dynamics, and discount sensitivity to anchor price ranges.
  - Modeled **hedging strategies under multiple scenarios** using Monte Carlo simulation (10,000 paths) to quantify risk/return trade-offs across delta-hedged and unhedged positions.
  - Built KMeans clustering to identify peer cohorts for benchmarking; integrated real-time market feeds to keep valuation assumptions current through the issuance window.

- **Bitcoin Time-Series Forecasting System**
  - Designed an **LSTM-based time-series forecasting model** on high-frequency price data — engineering lag features, rolling volatility, and order-book signals to improve directional accuracy by 18%.
  - Deployed real-time prediction API on AWS (EC2 + Lambda) with CloudWatch monitoring; built a Streamlit front-end for live signal visualization.

Education

- **Avila University** Kansas City, Missouri
  Master of Science in Computer Science — Quantitative Methods
  Coursework: Statistical Modeling, Mathematical Optimization, Financial Data Analysis, Machine Learning, Operations Research

- **JNTU** India
  Bachelor of Technology in Computer Science
  Coursework: Probability & Statistics, Linear Algebra, Numerical Methods, Applied Mathematics

Technical Skills

- **Quantitative:** Financial Engineering, Monte Carlo Simulation, Hedging Strategy Modeling, Sensitivity Analysis, Time-Series Forecasting, Factor Attribution, Scenario Analysis
  Finance: Equity Valuation, Comparable Company Analysis, Capital Markets, M&A Analytics, Debt & Equity Issuance, IPO Pricing, Client Reporting
  Programming: Python (advanced), SQL (advanced), Java (familiar), Q/KDB (familiar)
  ML & Data: XGBoost, Scikit-learn, PyTorch, FAISS, Pandas, NumPy, ETL Pipelines, PostgreSQL, MySQL
  Tools: AWS (EC2, Lambda), Docker, FastAPI, Power BI, Streamlit, Git